

ROLL NO:250701573

EXPERIMENT NO: 9 EXPERIMENT TO UNDERSTAND LOGISTIC REGRESSION FOR A DATASET

Program:

```
import numpy as np
import pandas as pd

df = pd.read_csv('Social_Network_Ads_modified.csv')
df
df.head()
features = df.iloc[:, [2, 3]].values
label = df.iloc[:, 4].values
```

Features

Label

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=37
)
model = LogisticRegression()

model.fit(x_train, y_train)
) train_score = model.score(x_train, y_train)
test_score = model.score(x_test, y_test)

print("Train Accuracy:", train_score)
```

```
print("Test Accuracy:", test_score)

from sklearn.metrics import classification_report

print(classification_report(label, model.predict(features)))
```

OUTPUT:

User ID	Gender	Age	EstimatedSalary	Purchased	
0	17000001	Male	41	74570	1
1	17000002	Female	28	19015	0
2	17000003	Male	25	79813	0
3	17000004	Male	53	45712	0
4	17000005	Male	55	118497	1
...	...	...	...	...	...
395	17000396	Female	22	57986	1
396	17000397	Female	42	103595	1
397	17000398	Female	19	119358	1
398	17000399	Female	27	82929	1
399	17000400	Male	47	91506	1

400 rows × 5 columns

User ID	Gender	Age	EstimatedSalary	Purchased	
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1	17000002	Female	28	19015	0
2	17000003	Male	25	79813	0
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4	17000005	Male	55	118497	1

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```

```

x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=37
)

```

Train Accuracy: 0.7375

Test Accuracy: 0.675

precision	recall	f1-score	support	
0	0.63	0.29	0.40	125
1	0.74	0.92	0.82	275
accuracy			0.72	400
macro avg	0.69	0.61	0.61	400
weighted avg	0.71	0.72	0.69	400